

# RAYMOND XIONG

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## EDUCATION

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**University of Massachusetts Amherst (UMass Amherst)**

*MS/PhD in Computer Science*

Aug. 2026 – May 2031

*Amherst, MA*

**Duke University**

*BS in Computer Science, BS in Statistical Science, Minor in Linguistics, magna cum laude*

Aug. 2022 – May 2026

*Durham, NC*

- GPA: 3.98 / 4.00; Graduation with the Highest Distinction in Computer Science; Graduation with Distinction Outside the Major in Linguistics
- Honors Thesis (advised by Monica Agrawal): Exploring False Assumptions and Incorrect Reasoning in Patient Information Search

## SELECTED PUBLICATION

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**Publication types:** J = Journal Article; C = Conference Paper; W = Workshop Paper; PR = Preprint.

\* = Co-first authorship.

[PR2] Lu, J., Yu, H., **Xiong, R. M.**, Zhang, A., Zhuo, D. (2026). Iteration Without Elaboration: A Simple ReAct Architecture Suffices for Text-to-SQL Generation. *arXiv*.

[C3] **Xiong, R. M.**, Chen, P., Dong, T., Lu, J., Goldstein, B., Zhuo, D., & Zhang, A. R. (2026). Reliable Curation of EHR Dataset via Large Language Models under Environmental Constraints. *Proceedings of the 14th IEEE International Conference on Healthcare Informatics*, 833-842.

[C2] **Xiong, R.**, Chen, Y. (2026). A Mixed-Effects Analysis of Addressee Honorifics in Japanese Voice Actor Livestreams. *Proceedings of the Linguistic Society of America*, 11 (1), 6052.

[PR1] **Xiong, R.**, Jia, F., Wong, L., Agrawal, M. (2026). What Patients Really Ask: Exploring the Effect of False Assumptions in Patient Information Seeking. *arXiv*.

[C1] Wong, L., Ali, A., **Xiong, R. M.**, Shen, S. Z., Kim, Y., & Agrawal, M. (2025). Position: Retrieval-augmented systems can be dangerous medical communicators. *Proceedings of the 42nd International Conference on Machine Learning, Proceedings of Machine Learning Research (PMLR)* 267, 82347-82359.

[J2] Xu, Q. \*, **Xiong, R. M.** \*, Zhao, M., & Wang, H. (2025). The structure, function, and quality of the social convoy for improvements in depressive symptoms in urban and rural China: A 3-year longitudinal cohort study. *International Psychogeriatrics*, 37(1), 100004.

[J1] **Xiong, R. M.**, Xie, T., Zhang, H., Li, T., Gong, G., Yu, X., & He, Y. (2022). The pattern of cortical thickness underlying disruptive behaviors in Alzheimer's disease. *Psychoradiology*, 2(3), 113–120.

## TALKS & PRESENTATIONS

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**Presentation types:** I = Invited Talk; O = Oral Presentation; P = Poster Presentation.

[I1] **Xiong, R.** (2026, January). Toward Trustworthy Language Technologies for Health Applications. BioNLP Lab, University of Massachusetts Amherst.

[O2] **Xiong, R.** (2026, January). A Mixed-Effects Analysis of Addressee Honorifics in Japanese Voice Actor Events. Linguistic Society of America Annual Meeting.

[P1] **Xiong, R.** (2025, December). What Patients Really Ask: Exploring the Effect of False Assumptions in Patient Information Seeking. Machine Learning for Health Symposium.

[O1] **Xiong, R.** (2025, October). A Mixed-Effects Analysis of Addressee Honorifics in Japanese Voice Actor Events. Duke University × Ochanomizu University Student Workshop, Duke University & Ochanomizu University.

## RESEARCH EXPERIENCE

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### UMass Amherst BioNLP Lab

Aug. 2026 – Present

*PhD student researcher*

*Amherst, MA*

#### **SUD Clinical Note Generation** | *advised by Hong Yu*

- Using multi-agent frameworks to generate realistic clinical notes for substance use disorder (SUD) patients.

### Duke University Agrawal Lab

Nov. 2024 – May 2026

*Undergraduate researcher*

*Durham, NC*

#### **Response Verifiability of LLM-powered RAG** | *advised by Monica Agrawal*

- Analyzed Google AI Overview & Perplexity responses to medical questions, to investigate response verifiability deficits in current-state large language model(LLM)-powered search engine systems in medical contexts. Built Playwright-based web scraping and LLM-as-a-judge automatic evaluation pipelines.

#### **False Assumptions in Patient Information Seeking** | *advised by Monica Agrawal*

- Investigated false assumptions in health-related questions that people commonly ask and curated a corresponding dataset. Tested various open-source and production LLMs to find that they struggle to identify incorrect assumptions in everyday questions.

### Duke University Chen Group

Mar. 2025 – May 2026

*Undergraduate researcher*

*Durham, NC*

#### **Japanese Sociopragmatics Analysis** | *advised by Yunchuan Chen*

- Analyzed the strategic use of addressee honorifics by Japanese native speakers in a YouTube series. Annotated data using automatic speech recognition models. Used generalized linear mixed-effects models to model the occurrence of addressee honorifics and investigate the role of conversation contextual factors.

### Duke University Zhang & Zhuo Lab

Mar. 2024 – May 2026

*Undergraduate researcher*

*Durham, NC*

#### **LLM-powered EHR Dataset Curation** | *advised by Anru Zhang and Danyang Zhuo*

- Created an end-to-end AI app to facilitate electronic health records data analytics for healthcare researchers. Improved execution accuracy of Text-to-SQL through few-shot and chain-of-thought prompting with schema information.

### Peking Uni Institute of Mental Health Dementia Care and Research Center

Jun. 2023 – Jun. 2024

*Intern researcher*

*Remote*

#### **Social Convoy & Depression** | *advised by Huali Wang and Mei Zhao*

- Investigated the influence of the social convoy and urban-rural status on the improvement of depression. Used Python to process data from a large-scale health database, perform logistic regression analysis, and create visualizations.

### Beijing Normal Uni Lab of Cognitive Neuroscience and Learning

Nov. 2019 – Sept. 2022

*Visiting high school student researcher*

*Beijing, CHN*

#### **Cortical Patterns of AD Disruptive Behaviors** | *advised by Yong He*

- Proposed a study examining the relationship between cortical thickness and disruptive behaviors in Alzheimer's disease patients. Secured funding from the China High School Science Talent Program. Processed MRI T1-weighted image data using the CIVET pipeline and conducted statistical testing in MATLAB.

## TEACHING EXPERIENCE

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### UMass Amherst

Aug. 2026 – Present

*Graduate teaching assistant*

*Amherst, MA*

#### **COMPSCI383 Artificial Intelligence**

Fall 2026

### Duke University

Jan. 2024 – May 2026

*Undergraduate teaching assistant*

*Durham, NC*

#### **COMPSCI572 Natural Language Processing**

Spring 2026, Spring 2025

- Developed and validated autograding infrastructure for CRF/Transformer implementation assignments.
- Collaborated with course staff to experimentally calibrate expected performance thresholds for a fine-tuning assignment.

#### **COMPSCI330 Design & Analysis of Algorithms**

Fall 2025, Spring 2025

#### **STA432 Stat Inference & Learning**

Fall 2024

#### **COMPSCI210 Intro to Computer Systems**

Fall 2024, Spring 2024

## OTHER EXPERIENCE

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### **Duke University Cogan Lab**

Sept. 2023 – May 2025

*Undergraduate intern technician*

*Durham, NC*

- Revamped intracranial electroencephalography data processing pipelines from Python to C to enhance efficiency.
- Led the packaging and publishing of the pipelines as the IEEG package on PyPI.

### **Duke Chinese Student Association**

Oct. 2022 – Apr. 2025

*President, Cultural VP & Chair (-Mar. 2024)*

*Durham, NC*

- Steered cultural integration and enhanced the impact of the community on campus by organizing cultural and social events with an average attendance of 100+; advertised through multiple channels to attract more non-Asian attendees than in previous years.
- Increased club funding by 60% during presidency.
- Initiated to create a web application collecting student course evaluation feedback to promote community support.

### **Duke Statistical Science Majors Union**

May 2023 – May 2024

*Communications officer*

*Durham, NC*

- Organized book-bagging information sessions, lunches with faculty and alum panels to strengthen academic and social bonds within the undergraduate statistical science community.

## SELECTED AWARDS & HONORS

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**Honorable Mention in the 2026 US NSF GRFP**

Apr. 2026

**Duke University Graduation with the Highest Distinction in Computer Science**

May 2026

**Duke University Graduation with Distinction Outside the Major in Linguistics**

May 2026

**Mathematical Contest in Modeling 2023 Meritorious Winner**

Apr. 2023

## RESEARCH SUPPORT

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**Duke CS Department Travel Award**

Nov. 2025

**Duke Linguistics Department Travel Award**

Oct. 2025

**Duke Asian/Pacific Studies Institute Conference Travel Award**

Sept. 2025

**Duke URS Conference Award**

Sept. & May 2025

**Duke URS Assistantship Award**

Sept. & May 2025

**Duke CS+ Program**

May 2024